

SDRAM Controller — Project Flow (V1 → V2)

V1 — Baseline (RTL → GDSII)

RTL

- Address decoder
- Bank manager
- Initialization FSM
- Read path
- SDRAM timing (tRCD, tRP, tRAS, tRC, CAS latency, tREFI)

Flow

- RTL verification
- ASIC synthesis
- Floorplan
- Placement
- CTS
- Routing
- GDSII

Status: complete tape-out-shaped flow. Strong standalone educational project.

V2 — Upgrades

Functional features

- ☐ Write support (V1 is read-focused)
- ☐ Auto-refresh controller
- ☐ Refresh scheduler
- ☐ Burst read support
- ☐ Burst write support
- ☐ Multiple outstanding requests
- ☐ Request queue / FIFO
- ☐ Error detection and reporting
- ☐ Power-down mode
- ☐ Self-refresh mode

Interface

- ☐ AXI4-Lite
- ☐ AXI4 (full — bursts, IDs, out-of-order)
- ☐ APB
- ☐ Wishbone

Verification

- ☐ SystemVerilog testbench
- ☐ Assertions (SVA)
- ☐ Functional coverage
- ☐ UVM-based verification (later, optional)

Physical design

- ☐ Sky130 standard-cell implementation
 - ☐ Static Timing Analysis
 - ☐ Power estimation
 - ☐ DRC / LVS
 - ☐ Better floorplanning
 - ☐ Area optimization
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Suggested ordering notes

1. **Write path before refresh.** Refresh logic has to arbitrate against pending writes; building it against a read-only datapath means rewriting the arbiter.
2. **Refresh before bursts.** Burst state machines get much harder to interrupt correctly once refresh can preempt mid-transfer — better to have refresh in place first.
3. **FIFO / outstanding requests before AXI4-full.** AXI4 bursts and multiple IDs assume a queue underneath; AXI4-Lite is the safe first interface.
4. **SVA early, not last.** Timing-parameter assertions (tRCD/tRP/tRAS violations) are cheapest to write while the timing logic is fresh, and they pay for themselves on every later change.
5. **STA + power after each functional milestone,** not once at the end — area and timing regressions are easier to attribute to a single feature.